MERLIN STUWART

AI/ML & Full-Stack Software Developer | Python · React.js · RAG Systems · LLMs

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PROFESSIONAL SUMMARY

AI/ML and Full-Stack Developer with 1.5+ years of hands-on experience building production applications and AI-powered systems using Python, FastAPI, React.js, LangChain, REST APIs, MySQL, and MongoDB. Delivered a client-facing WhatsApp automation bot that reduced manual agent response time by ~40%, and built RAG pipelines with embeddings and semantic search, plus fine-tuned LLMs for domain-specific Q&A systems. Skilled in model deployment, CI/CD, and secure LLM integration.

KEY SKILLS

Programming Languages: Python, JavaScript, TypeScript.

LLM & Generative AI: LLM Integration, Retrieval-Augmented Generation (RAG), RAG Evaluation, LLM Fine-Tuning, Prompt Engineering, Embeddings, Semantic Search, NLP, Machine Learning, AI Security

AI Frameworks & APIs: LangChain, LangGraph, OpenAI API, Hugging Face

Vector Databases: ChromaDB, FAISS.

Backend & Web: FastAPI, REST API Development, ETL Pipelines, Celery, Kafka, Redpanda, React.js, Angular, HTML5, CSS3, Electron

Databases: MySQL, PostgreSQL, MongoDB, Redis, ClickHouse

MLOps & DevOps: Model Deployment, CI/CD, Docker, Docker Compose, GitHub Actions, Git/GitHub, n8n Workflow Automation, WhatsApp Business API.

Practices: Object-Oriented Programming (OOP), API Integration, Unit Testing, Debugging, Version Control.

WORK EXPERIENCE

Junior Software Developer — Wyzmindz Solutions Pvt Ltd *Mar 2025 – Present*

- Independently designed, built, and deployed a client-facing WhatsApp automation bot as sole project owner, from requirements to production rollout, with strong client satisfaction.
- Integrated LLMs and RAG pipelines (embeddings, vector databases, semantic search) into business applications, collaborating with cross-functional engineering teams.
- Developed full-stack features using Python, FastAPI, React.js, REST APIs, and MySQL/MongoDB across multiple live company projects.
- Containerized applications with Docker and used CI/CD practices for consistent, repeatable deployments.

Project Management Intern — Salesforce Info Tech *2023 (1 Month)*

- Supported tracking of Salesforce-based project deliverables; gained exposure to CRM workflows, documentation, and cross-team coordination.

PROJECTS

Advaita ETL — Enterprise Data Integration & Automation Platform

Tech: Python, FastAPI, Angular, TypeScript, MongoDB, MySQL, PostgreSQL, Redis, Celery, Kafka, Redpanda, ClickHouse, Docker, GitHub Actions

- Developed and enhanced an enterprise ETL platform that integrates multiple data sources and automates extraction, transformation, and loading through configurable, node-based pipelines.
- Built REST APIs with Python, FastAPI, and MongoDB for pipeline management, data connections, and job execution; implemented connectors for MySQL, PostgreSQL, MongoDB, CSV, Excel, REST APIs, Kafka, and Redpanda.
- Implemented asynchronous processing and scheduled pipeline execution with Celery, Celery Beat, and Redis; contributed to job monitoring and multi-tenant architecture.
- Developed Angular frontend modules for visual pipeline building, data mapping, transformations, and dashboards; used Docker, Docker Compose, and GitHub Actions for containerized development and CI.

Autosherpa — WhatsApp Business Automation Bot

Tech: Python, WhatsApp Business API, Docker

- Built a Docker-containerized WhatsApp bot handling leads, bookings, and support workflows, reducing manual agent response time by ~40%.
- Integrated AI-driven intent classification to cut first-response time from minutes to seconds.

Autosherpa — WhatsApp Automation Desktop Application

Tech: Electron, JavaScript, Python

- Built an Electron desktop control app that lets non-technical staff start/stop bot sessions and monitor multiple WhatsApp automation instances in real time, reducing manual oversight.

Suryajosvi Reflection Tracker

Tech: React.js, Python, Machine Learning, RAG, Embeddings, Vector Database.

- Built a full-stack AI-powered reflection tracker with React.js and a RAG pipeline backed by a vector database, delivering personalized insights from users' journal entries.

NEET Preparation Assistant — Fine-Tuned LLM

Tech: Python, LLM Fine-Tuning, NLP

- Fine-tuned a custom LLM on NEET syllabus content (Physics, Chemistry, Biology) to build a domain-specific Q&A and doubt-solving assistant, reducing reliance on generic AI tools for exam preparation.

EDUCATION

B.E. Computer Science Engineering — Aarupadi Veedu Institute of Technology, Chennai *2019 – 2023*

CERTIFICATIONS & TRAINING

Python Full Stack Development — JSpiders (2023–2024) | Microsoft Power BI | n8n Yellow Belt — Workflow Automation